

OpenCode Master Implementation Prompt

A self-contained execution prompt for the existing repository.

VERSION	STATUS	TARGET
v0.3	Feed directly into OpenCode	OpenCode / implementation team

Usage: Give OpenCode this entire document together with repository access. It should inspect before editing, preserve working sensing code, fix the iOS camera and experiment architecture first, and report what remains unverified on physical hardware.

1. ROLE AND FIRST ACTION

You are working on the existing EmOcean AI web application. Do not rebuild blindly from scratch. First inspect the complete repository, architecture, dependencies, routes, camera lifecycle, current physiological/emotion implementation, experiment state, styling system, tests, deployment configuration, and unfinished code.

Before editing, return: (1) current architecture; (2) likely reason camera fails on iOS; (3) which requested features already exist; (4) which current implementations should be preserved; (5) concrete file-by-file implementation plan.

2. PRODUCT OBJECTIVE

The product is EmOcean AI (target domain: emocean.ai). It is a privacy-first browser experience that uses the user's front-facing camera to estimate observable physiological and facial-response signals while controlled digital environments are shown. The first consumer use case is to identify colors and chat themes associated with the user's calmest and positively activating response, then export a practical theme.

This is an experimental wellness/personalization product, not a diagnostic or medical device.

3. PERSISTENT EXPERIENCE STATE MACHINE

Implement the experience as one persistent browser route/state machine: CONSENT -> CAMERA_SETUP -> SIGNAL_ACQUISITION -> BASELINE -> COLOR_SCREENING_A -> COLOR_SCREENING_B -> COLOR_FINALISTS -> CHAT_THEME -> OPTIONAL_FONT -> OPTIONAL_EMOJI -> SUBJECTIVE_CHOICE -> RESULTS -> EXPORT.

Do not navigate between separate routes while the camera is active. The MediaStream and video element should remain mounted continuously from permission through experiment completion.

4. P0 - FIX IOS SAFARI CAMERA

Diagnose and fix the current real-iPhone failure before cosmetic work is considered complete. Requirements: camera request originates from an explicit user tap; HTTPS; navigator.mediaDevices.getUserMedia; prefer { video: { facingMode: "user" }, audio: false }; avoid facingMode exact unless necessary; attach through video.srcObject; video uses autoplay, muted and playsInline; explicitly call play() after metadata if needed; do not unmount the video across stage changes; add visibilitychange/pageshow recovery and track ended/mute/unmute handling; add readable errors for permission denied, unavailable camera, stalled stream, unsupported browser, model load failure and insufficient light; verify on a physical iPhone Safari device.

5. CAMERA PREVIEW

Once permission is granted, always show a mirrored camera preview. Keep it visible during signal acquisition, baseline, color trials, finalist trials, chat-theme trials, font trials and emoji trials. Use a small floating preview during full-screen stimuli. Show face detected, signal quality, movement warning and lighting warning. During actual experimental windows, do not show continuously changing HR/breathing numbers.

6. PRIVACY CONSENT

Before requesting camera permission show: "Your camera is analyzed on this device to estimate physiological and facial-response signals. Raw video is not recorded or uploaded. Closing this page ends camera access. Results are for personal exploration and are not medical measurements." Require explicit confirmation. Do not upload raw video. Do not upload raw biometric data by default. Any future upload of derived physiological metrics requires separate explicit informed consent.

7. SENSOR ARCHITECTURE

Create or preserve a clean SensorHub abstraction with replaceable adapters/modules for rPPG, heart rate, BVP, beat intervals, signal quality, PRV outputs when valid, respiration estimate, facial expression, face detection, head pose, eye state, gaze, speaking/jaw movement and movement/stability. Prefer Vital Camera SDK / FacePhys as the primary browser rPPG implementation if compatible. Pin reviewed versions/commits; do not depend on an unpinned GitHub main branch in production. Self-host model/WASM assets when practical. Never fabricate physiology when unavailable. Simulation mode may exist only behind an explicit development flag.

8. LIVE WOW SCREEN

Before the experiment, show a compelling but defensible live screen: heart rate BPM; breathing estimate breaths/min; pulse waveform; camera-derived pulse variability only with enough clean beats; facial-response tendency; motion/stability; signal quality. Do not make arbitrary pseudo-precise Calm/Focus/Curious numbers primary outputs unless a validated model explicitly supports them. Facial-expression models indicate observable facial responses, not objective knowledge of internal emotion.

9. BASELINE

After adequate signal acquisition, run approximately 25-30 seconds of baseline. User copy: "Look comfortably at the screen. Breathe normally. Blink normally. Try not to talk or move too much." Do not instruct slow breathing. Do not show paced breathing. Do not ask the user to look away. Keep the camera visible and use a physiologically neutral ocean-ripple progress experience with messages: Finding your face; Checking the light; Finding your pulse; Learning your normal rhythm; Almost there.

10. COLOR EXPERIMENT

Remove the sequential 30-second-per-color implementation. Use seeded randomized repeated trials. Start with five controlled colors stored separately from brand colors. Use a perceptual color space such as OKLCH and approximately match lightness. Screening A: randomize the five colors; each stimulus about 12 seconds; full-screen/near-full-screen; first about 3 seconds marked transition and excluded; about 4 seconds neutral washout between stimuli. Screening B: repeat the same five colors in a new random order. Persist randomization seed and exact order. Finalists: select top two valid candidates and retest each for about 30 seconds in randomized order with neutral washout. If signal quality or repeat agreement is insufficient, return an uncertain result rather than forcing a winner.

11. FULL SCREEN + OPTICAL CONFOUND

Experimental color must visually dominate the viewport; do not use small swatches/cards. Keep the floating camera/signal card visible. Because display color can alter light reflected from the face and contaminate rPPG, match perceptual lightness, log exact stimulus/timing, log relevant frame luminance/quality, exclude transition windows, keep device brightness stable where possible, warn on large illumination change, and keep correction logic modular for later contact-PPG validation.

12. CHAT TIDES

After color screening, turn the best candidate colors into accessible realistic chat themes. A theme includes background, incoming bubble, outgoing bubble, text, accent and timestamps. Use a real messaging layout, typing/message timing and full background changes. Do not simply show static color swatches. Ensure readable contrast.

13. NEUTRAL CHAT CONTENT

Create multiple emotionally neutral message sets with similar word count, reading difficulty, number of bubbles, timing and visual density. Randomize message-set assignment independently from theme condition so content cannot systematically confound color.

14. TYPOGRAPHY

Test typography separately from color when possible. Candidate categories may include System Sans, Humanist Sans, Rounded Sans, Soft Serif and Monospace. Keep color constant during initial font trials; later test finalist font + finalist color combinations. Do not promise arbitrary custom font installation inside WhatsApp when unsupported.

15. EMOJI MIRROR

Keep emoji trials separate from color/font trials. Candidate set may include smile, bright smile, relieved/calm, neutral and concerned expressions. Hold color/font/chat content constant. Measure calibrated facial-expression outputs plus HR, respiration and movement. Do not instruct the user to imitate the emoji. Describe results as spontaneous facial-response/mimicry tendencies, not proof of internal emotion.

16. NO BIOFEEDBACK DURING TRIALS

Before the experiment, live HR and breathing are useful. During experimental windows hide exact HR, respiration, PRV and calm-score values. Show only Measuring, Signal good, Hold still, Improve lighting and progress. This reduces conscious physiological manipulation.

17. SUBJECTIVE CHOICE

After physiology trials, ask which visual environment consciously felt best. Keep the choice separate from the physiological ranking. The result may say: "You chose Lavender. Your strongest calm-associated physiological response was Sea Glass."

18. RESULTS AND CONFIDENCE

Create a polished result screen with Calm Current, Activation Current and a stress-like result only when several independent signals agree. Example: SEA GLASS #BBD3C6; heart rate -4.1 BPM vs session baseline; breathing -1.6/min; movement -12%; repeat agreement 2/2; confidence 84%. Confidence should include signal-quality coverage, valid duration, face-detection coverage, movement, illumination stability, repeat agreement, winner margin and cross-signal agreement. If insufficient: "We didn't get a clear winner this time." Offer retry. Never manufacture certainty.

19. EXPORT

Generate locally: wallpaper PNG, HEX, OKLCH, recommended incoming/outgoing bubble colors, text color, theme JSON and a readable result summary. Provide practical instructions for recreating the appearance in WhatsApp and Telegram.

20. BRANDING

Replace Mirror/01 and Coloring AI consumer branding with EmOcean AI. Primary proposition: "Your screen affects you. Let's find out how." Alternative short tagline: "See what moves you." Supporting copy: "EmOcean AI uses your camera to observe subtle changes in your pulse, breathing and facial response while you experience different digital environments." Primary CTA: "Find my colors". Trust chips: No wearable; Runs on your device; Video never uploaded; About 5-7 minutes. Visual tone: curious, optimistic, premium, approachable, slightly playful, ocean-inspired, scientifically credible, non-clinical and non-psychedelic.

21. BRAND PALETTE

Use Deep Ocean #103438; Lagoon #1CB7A1; Aqua #61D6E5; Coral #FF7D73; Sun #FFC85A; Lavender #A894F4; Foam #F6FBF9 for the brand. These are not automatically the experimental stimuli. Keep brand and experiment palettes separate in code.

22. MOBILE-FIRST

Support 320 px, 375/390 px iPhones, 430 px large phones, common Android, tablet and desktop. No horizontal overflow. Respect safe-area insets. Avoid controls under notch/Dynamic Island/browser chrome. Use touch-friendly sizes. Camera preview cannot obscure instructions. Use 100dvh rather than naive 100vh where appropriate.

23. PERFORMANCE

Lazy-load heavy sensing models only after the user clicks Start. The marketing page must not download physiological models. Keep camera processing off the main UI thread when possible; use supported Workers. Show real model-loading progress and add timeout/error recovery.

24. TESTING

Add unit tests for seeded randomization, complete repeats, transition exclusion, low-SQL rejection, uncertain result, camera state preservation, accessible chat contrast and export generation. Add Playwright flows for landing->consent, camera mocked flow, calibration, color experiment, results, mobile viewport, desktop viewport, denied camera and sensor initialization failure. Create a manual physical-device QA checklist including iPhone Safari.

25. DEVELOPMENT PROCESS

Preserve working sensing algorithms. Keep sensing, experiment, UI and export layers cleanly separated. Make changes in logical commits. After implementation, run tests and report exactly what was tested, what passed, what remains unverified on physical hardware, and which scientific assumptions still require external validation.

Reference URLs for implementation

Live app: <https://emo-ocean-neon.vercel.app>

Vital Camera SDK: <https://github.com/KegangWangCCNU/vitalcamera-sdk>

FacePhys demo: <https://github.com/KegangWangCCNU/FacePhys-Demo>

MediaPipe Face Landmarker: https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker/web_js

MDN getUserMedia: <https://developer.mozilla.org/en-US/docs/Web/API/MediaDevices/getUserMedia>

OpenCode: <https://opencode.ai/docs/>

Final instruction to OpenCode: Do not optimize for speed at the expense of measurement integrity. Fix P0 camera reliability and experimental state logic before declaring the redesign complete.